

MORGAN STANLEY  
700 WELLINGTON ST.  
MONTREAL, QC, CANADA H3C 3S4

**Internship Application for Summer 2026 Quantitative Finance Program in Montreal**

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To whom it may concern,

**About Me**

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Thank you for considering my application. I am currently a PhD candidate in experimental quantum optics at the University of Toronto under Prof. Joseph Thywissen. My interest in financial markets developed early in my PhD during the post-COVID-19 period of heightened market volatility and intense focus on interest-rate policy. Witnessing the economic dynamism of that period inspired me to pursue quantitative finance as a career path, especially in areas related to Fixed Income or macroeconomics. As I approach completion of my program, I am eager to apply my rigorous training in research, data analysis, and statistical modeling to an exciting and innovative field.

**Why Morgan Stanley?**

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I am drawn to Morgan Stanley for its leadership in the industry and its reputation for excellence. I first learned about the Summer Internship Program through a senior graduate student in my research group, whose highly positive experience ultimately led to a full-time role at the firm. His endorsement played a significant role in my decision to apply, and it highlighted to me the strength of Morgan Stanley's culture and the quality of its training.

**Why Me?**

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I believe my experience as an experimental physicist has prepared me well for quantitative finance. My research requires a multi-disciplinary skill set that emphasizes analytical thinking, adaptability, and rapid problem-solving. I look forward to the opportunity to contribute my skills at Morgan Stanley this summer and to learn from the firm's talented professionals.

Sincerely,

**Kevin Xie**

*Attached: Résumé*

# Kevin Xie

PHD CANDIDATE · PHYSICS · UNIVERSITY OF TORONTO

75 Kruger Road, Markham, ON, Canada, L3S 3Y8

☎ (+1) 416-993-6322 | ✉ kevin.gs.xie@gmail.com | 🏠 Kevin G. S. Xie

## Summary

PhD candidate in experimental quantum optics with 8+ years of experience spanning physics research, engineering, and industry R&D. Strong foundation in quantitative analysis, modeling, and development. Motivated by learning, adapting, and applying technical skills to new domains, including finance.

## Experience

### University of Toronto (Prof. Joseph Thywissen)

Toronto, Canada

PHD RESEARCHER, EXPERIMENTAL QUANTUM OPTICS

May 2020 – Present

- **Research focus:** precision measurements, high-dimensional data analysis, control of complex physical systems.
- Conducted experiments on ultracold quantum gases, including spectroscopy of strongly interacting fermions to study many-body phenomena.
- Applied model fitting, parameter estimation, and uncertainty analysis to extract physical parameters from noisy data.
- Deployed and maintained custom control, real-time monitoring, and automated data-acquisition systems (Python, C++).
- Led a research team of four, directing experimental strategy, coordinating measurement protocols, and managing data-analysis pipelines.
- Regularly communicated results to international collaborators; presented findings at conferences and peer-reviewed journals.

### University of Waterloo (Prof. Jan Kycia)

Waterloo, Canada

UNDERGRADUATE RESEARCHER

Jan. – Sep. 2017, Sep. 2019 – Apr. 2020

- Engineered and analyzed thermal properties of components for a hybrid Helium-3 dilution refrigerator aimed at millikelvin-scale spin-qubit characterization.

### Lumentum

Ottawa, Canada

OPTICAL ALGORITHMS DEVELOPER

Sep. – Dec. 2018, May – Aug. 2019

- Built a predictive failure-detection neural network model (>96% accuracy) to optimize calibration of LCoS components in optical networking devices, aimed at improving throughput and automation.
- Standardized test-station infrastructure across multiple facilities, eliminating bottlenecks and increasing throughput by up to 10%.
- Conducted comprehensive performance studies on next-generation bandwidth-enhancement algorithms for optical module R&D.
- Replaced outdated internal component data retrieval tool with modern Python-based solution adopted company-wide.

### TRIUMF (T2K Canada)

Vancouver, Canada

UNDERGRADUATE RESEARCHER

Sep. 2017 – Apr. 2018

- Detected and corrected periodic temperature fluctuations that introduced systematic error in photomultiplier-tube characterization. Validated and reproduced by international collaborators (KEK).

## Extracurricular Activities

### Kaggle, Jane Street Group

Online

COMPETITION ENTRANT

2024

- Applied quantitative finance principles and machine learning algorithms (PyTorch, TensorFlow) to large-scale financial time-series data; developed feature engineering and model optimization with high-performance data tools (Polars).

### University of Toronto Graduate Students' Union

Toronto, Canada

MEMBER OF BOARD OF DIRECTORS AND ELECTIONS & REFERENDA COMMITTEE

Oct. 2023 – Apr. 2024

- Organized and supported policy development and governance initiatives affecting 20,000+ graduate students, contributing to improvements in academic, financial, and community resources.

### The 27th International Conference on Atomic Physics (ICAP 2022)

Toronto, Canada

VOLUNTEER

July 2022

- Assisted in organizing and running an international conference with 500+ participants, providing on-site logistics, A/V support, and management of virtual conference components.

## Skills

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|--------------------|-----------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python (NumPy, SciPy, Pandas), MATLAB, Mathematica, C++, Linux, Git, LaTeX                    |
| <b>Data</b>        | Uncertainty Analysis, Optimization, Signal Processing, Machine Learning (PyTorch, TensorFlow) |
| <b>Languages</b>   | English (native), Cantonese (beginner/intermediate)                                           |
| <b>Interests</b>   | Music (Jazz), Boxing, Geopolitics, History                                                    |

## Education

### University of Toronto

PHD PHYSICS

*Toronto, Canada*

*Sep. 2020 - Present*

### University of Waterloo

BSC PHYSICS, CO-OP

*Waterloo, Canada*

*Sep 2015 - Apr 2020*

## Publications

\* indicates shared first authorship.

### PUBLISHED

- C. J. Dale\*, **K. G. S. Xie\***, K. Pond Grehan, S. Zhang, J. Maki, and J. H. Thywissen. Emergent s-wave interactions in orbitally active quasi-two-dimensional Fermi gases. *Phys. Rev. A* 110, L051302 (2024)
- K. G. Jackson, C. J. Dale, J. Maki, **K. G. S. Xie**, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent s-wave interactions between identical fermions in quasi-one-dimensional geometries. *Phys. Rev. X* 13, 021013 (2023)
- C. R. C. Buhariwalla, Q. Ma, L. DeBeer-Schmitt, **K. G. S. Xie**, D. Pomaranski, J. Gaudet, T. J. Munsie, H. A. Dabkowska, J. B. Kycia, and B. D. Gaulin. Long-wavelength correlations in ferromagnetic titanate pyrochlores as revealed by small-angle neutron scattering. *Phys. Rev. B* 97, 224401 (2018)

### IN REVIEW

- K. G. S. Xie\***, C. J. Dale\*, K. Pond Grehan, M. F. Wang, T. Enss, P. S. Julianne, Z. Yu, and J. H. Thywissen. Dimer-projection contact and the clock shift of a unitary Fermi gas. (under review)

## Presentations

### CONTRIBUTED TALKS AND POSTERS

- K. G. S. Xie**, C. J. Dale, K. Pond Grehan, M. F. Wang, Z. Yu, P. S. Julianne, T. Enss, and J. H. Thywissen. Clock shift and contact of a unitary Fermi gas inferred by dimer association. Oral presentation: DAMOP 2025, Portland, Oregon, U.S.A.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent s-wave dimers near a  $p$ -wave Feshbach resonance in a strongly confined Fermi gas. Oral presentation and poster: ICPEAC 2023, Ottawa, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, S. Zhang, and J. H. Thywissen. Associating emergent s-wave dimers along strongly confined directions of a spin-polarized Fermi gas. Oral presentation: DAMOP 2023, Spokane, Washington, U.S.A.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Feshbach-enhanced odd- and even-wave interactions between identical fermions in quasi-one-dimensional confinement. Poster: ICAP 2022, Toronto, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Kokkelmans, S. Smale, P. S. Julianne, and J. H. Thywissen.  $P$ -wave interactions in a spin-polarized Fermi gas with and without confinement. Poster: DAMOP 2022, Orlando, Florida, U.S.A.

## Honors & Awards

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|------|------------------------------------------------------------------------|-------------------------|
| 2025 | <b>SGS Conference Travel Grant</b> , University of Toronto             | <i>Toronto, Canada</i>  |
| 2023 | <b>Ontario Graduate Scholarship</b> , University of Toronto            | <i>Toronto, Canada</i>  |
| 2021 | <b>Canada Graduate Research Scholarship – Master’s</b> , NSERC         | <i>Toronto, Canada</i>  |
| 2020 | <b>Dean’s Honours List</b> , University of Waterloo                    | <i>Waterloo, Canada</i> |
| 2017 | <b>Undergraduate Student Research Award</b> , NSERC                    | <i>Waterloo, Canada</i> |
| 2016 | <b>President’s Scholarship of Distinction</b> , University of Waterloo | <i>Waterloo, Canada</i> |

## Teaching Experience

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|------------|------------------------------------|------------------------|
| 2025       | <b>PHY324</b> , Teaching Assistant | <i>Toronto, Canada</i> |
| 2025       | <b>PHY293</b> , Teaching Assistant | <i>Toronto, Canada</i> |
| 2024       | <b>PHY294</b> , Teaching Assistant | <i>Toronto, Canada</i> |
| 2024       | <b>PHY405</b> , Teaching Assistant | <i>Toronto, Canada</i> |
| 2023       | <b>PHY424</b> , Teaching Assistant | <i>Toronto, Canada</i> |
| 2021, 2022 | <b>PHY224</b> , Teaching Assistant | <i>Toronto, Canada</i> |
| 2020, 2021 | <b>PHY131</b> , Teaching Assistant | <i>Toronto, Canada</i> |